Example Math Diary entry - Leon Pernak 2026-11-02

Explanation. Topic: Prime decomposition
Recall that a prime number is an integer divisible only by 1 and by itself. The prime decomposition (or factorization) theorem states that any integer can be written as a product of powers of prime numbers. Additionally, such a decomposition is unique up to the order of the factors (for example $14 = 7 \cdot 2 = 2 \cdot 7$ has prime factors 2 and 7, and is the only integer that can be written as a product of one 2 and one 7). The theorem is essential when comparing for example two integers: If we have an integer equation $2^n = 3^m$, we can see that the only solutions is $n=m=0$, because otherwise the prime factorizations on both sides .. would have different primes (2 vs 3).
The proof proceeds through Euclid's lemma (if a prime p divides a product $a \cdot b$, it must divide one of its factors), inductively separating prime factors from the number to decompose.
Some fun fact beyond the course: Prime factorization is computable ~~in polynomial time or exponential time~~ in many different ways, and it's an open problem whether the time complexity lies below ~~exponential~~ polynomial. Many cryptography algorithms rely on prime factorizations, a fast algorithm would likely break many encryption methods.

Lecture material: Pretty easy, I've seen most of it in previous lectures. Lemma 7.1 b) took me some time to understand.
Exercises: The exercises were pretty boring, can we do more on Euclid's algo
Learning: I read the book chapter and had a discussion with an AI chatbot on prime decomposition. I did not attend the lecture. I solved the exercises well, so I think I did well enough. Can we do more examples in the lecture?

POTW: The Chinese Remainder Theorem.
This is my pick of the week because in previous lectures, I never understood the proof and it was key for ~~the applicati~~ my understanding of congruence equations. What helped me particularly was the constructive proof, i.e that a solution to the equation is explicitly constructed. The proof then just verifies that it's indeed a solution. I have not understood so far why the different moduli need to be coprime, in my mind, the proof works perfectly w/o the assumption.